

王子涵 2023201781

房价：

--- 最终房价性能表 (包含 CV MAE) ---

	OLS	Lasso (alpha=0.0000)	Ridge (alpha=2.98)
MAE_in	450,208.97	450,217.46	450,213.73
RMSE_in	1,029,328.95	1,029,553.25	1,029,462.03
MAE_out	446,989.28	446,158.12	446,577.94
RMSE_out	1,033,547.67	1,028,958.69	1,031,012.27
MAE_cv	450,812.56	451,671.99	451,685.01

房租：

--- Best Rent ElasticNet v2 模型性能已记录 ---

	OLS	Lasso (alpha=0.0000)	Ridge (alpha=1.62)
MAE_in	121,805.46	121,814.74	121,812.34
RMSE_in	302,271.70	302,398.49	302,367.40
MAE_out	124,888.84	121,773.16	124,042.73
RMSE_out	292,710.00	293,741.15	292,782.73

OLS CV MAE: 122,062.05 Lasso CV MAE: 122,053.40 Ridge CV MAE: 122,052.48

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房价模型综合性能比较

	模型名称	样本内R ²	样本外R ²	交叉验证R ²	Kaggle得分	训练集RMSE	验证集RMSE	验证集MAE
Linear Regression	Ridge	0.8998	0.8983	0.8996	65.7	0.2633	0.2624	0.1988
		0.8998	0.8983	0.8996	65.1	0.2633	0.2624	0.1988
	Elastic Net	0.8997	0.8983	0.8996	65.8	0.2634	0.2625	0.1990
	Lasso	0.8993	0.8977	0.8992	65.7	0.2639	0.2633	0.1995

R²指标对比总结

	模型	样本内R ²	样本外R ²	交叉验证R ²	Kaggle得分
Linear Regression	Ridge	0.8998	0.8983	0.8996	65.7
		0.8998	0.8983	0.8996	65.1
	Elastic Net	0.8997	0.8983	0.8996	65.8
	Lasso	0.8993	0.8977	0.8992	65.7

🏆 最佳模型: Ridge
最佳验证集RMSE: 0.2624

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租金模型综合性能比较

	模型名称	样本内R ²	样本外R ²	交叉验证R ²	Kaggle得分	训练集RMSE	验证集RMSE	验证集MAE
Linear Regression	ElasticNet	0.8693	0.8664	0.8687	65.8	0.2746	0.2762	0.2012
	Ridge	0.8694	0.8658	0.8687	65.7	0.2745	0.2768	0.2016
	Lasso	0.8687	0.8651	0.8683	65.7	0.2753	0.2775	0.2024
	Linear Regression	0.8694	0.8622	0.8680	65.1	0.2745	0.2805	0.2045

R²指标对比总结

模型	样本内R ²	样本外R ²	交叉验证R ²	Kaggle得分
ElasticNet	0.8693	0.8664	0.8687	65.8
Ridge	0.8694	0.8658	0.8687	65.7
Lasso	0.8687	0.8651	0.8683	65.7
Linear Regression	0.8694	0.8622	0.8680	65.1

🏆 最佳模型: ElasticNet
验证集RMSE: 0.2762
验证集R²: 0.8664
验证集MAE: 0.2012

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	OLS	Lasso (alpha=0.0001)	Ridge (alpha=16.68) \
MAE_in	4.130239e+05	414791.843506	414666.445534
RMSE_in	9.383783e+05	948672.536536	948942.178520
MAE_out	NaN	412672.411985	412913.036844
RMSE_out	NaN	951565.625634	951937.820453
R ² _in	9.199472e-01	0.919151	0.919176
R ² _out	-3.570738e+05	0.917461	0.917461
R ² _cv	-1.128964e+25	0.917868	0.917840
ElasticNet (a=0.0001,l1=1.00)			
MAE_in		414791.843506	
RMSE_in		948672.536536	
MAE_out		412672.411985	
RMSE_out		951565.625634	
R ² _in		0.919151	
R ² _out		0.917461	
R ² _cv		0.917877	

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--- 房价版各模型 6 折 CV MAE 结果（基于全流程预处理数据） ---

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	6折CV MAE（原始房价）
OLS	N/A
Lasso (alpha=0.0001)	419,806.46
Ridge (alpha=16.68)	420,025.84
ElasticNet (a=0.0001,l1=1.00)	419,806.46

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===== 租金模型综合性能比较 =====

	OLS	Lasso (alpha=0.0005)	Ridge (alpha=42.81) \
样本内R ²	8.681000e-01	0.8679	0.8680
样本外R ²	8.570000e-01	0.8654	0.8657
交叉验证R ²	-1.830610e+24	0.8675	0.8674
MAE_in	1.218957e+05	121994.7164	121881.2152
RMSE_in	2.845959e+05	285374.9764	284520.8357
MAE_out	1.244461e+05	120925.7285	120798.8253
RMSE_out	2.770210e+05	278066.5054	277855.6068
MAE_cv	NaN	122245.2619	122245.9737

	ElasticNet (alpha=0.0007, l1=0.70)
样本内R ²	0.8679
样本外R ²	0.8654
交叉验证R ²	0.8675
MAE_in	121989.9205
RMSE_in	285334.7442
MAE_out	120925.5714
RMSE_out	278055.3227
MAE_cv	122239.7746

===== 最佳模型 =====

模型名称: Ridge (alpha=42.81)

样本外R²: 0.8657

验证集RMSE: 277855.6068

验证集MAE: 120798.8253

交叉验证R²: 0.8674

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房价模型性能比较表（原始价格尺度）：

	train_mae	val_mae	train_rmse	val_rmse
OLS	305101.92	309476.77	750982.53	769455.21
Lasso	331817.39	329361.33	788525.76	790593.39
Ridge	304292.31	307780.70	749395.94	761697.50
ElasticNet	313951.26	312276.21	760792.10	766322.78

最佳模型：Ridge

验证集MAE：307,780.70

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租金模型性能比较表（原始价格尺度）：

	train_mae	val_mae	train_rmse	val_rmse
OLS	145511.36	147528.13	343160.99	349614.97
Lasso	149424.47	151581.96	357337.71	364906.02
Ridge	144708.68	147257.92	342160.56	350215.35
ElasticNet	146650.36	148770.39	350187.92	358050.13

最佳模型：Ridge

验证集MAE：147,257.92